

Data Card — Optimizing RAG Document Retrieval for Agronomic Advice

Dataset Summary

Our dataset is a text-based agricultural information retrieval dataset for the project “Optimizing RAG Document Retrieval for Agronomic Advice.” It contains 695 agricultural documents, 308 training queries, 4,194 query-document relevance judgments, and 200 test queries provided through the Kaggle competition. The document text contains an average of 66.1 words per document, with document lengths ranging from 20 to 180 words. Training queries contain an average of 9.4 words, ranging from 6 to 16 words. The documents include fields such as document ID, title, text, source, crop, country, origin, source URL, and license. Relevance judgments use scores from 0 to 3, with 2,724 judgments at 0, 728 at 1, 186 at 2, and 556 at 3. The system receives an agricultural question as text and retrieves and ranks five documents from the corpus. This is a retrieval and ranking task rather than a conventional classification task. Retrieval quality is evaluated using nDCG@5, which considers both document relevance and ranking position.

Ethics and Representation

The project uses the dataset supplied through the Kaggle competition and does not involve collecting new personal information from individuals. The team is not conducting additional web scraping for this project. The provided document records contain source and license information, and the records inspected by the team include documents identified as synthetic and CC0.

Representation remains an important limitation. The 695-document corpus may not cover every agricultural region, crop, country, language, farming practice, or type of agricultural question. The relatively short document length, with an average of 66.1 words, also describes the particular corpus used in this competition and should not be assumed to represent the full range of agricultural knowledge available in real-world settings. Similarly, the 308 training queries, averaging 9.4 words, represent only a limited set of information needs.

The relevance labels are also unevenly distributed: 2,724 judgments are labelled 0, compared with 556 labelled 3. This distribution is important when interpreting evaluation results. The team plans to examine poorly performing queries and compare different retrieval approaches. Broader data and evaluation involving appropriate agricultural experts and intended users would be needed before real-world deployment.

Values and Community Impact

The dataset supports the project’s aim of improving access to relevant agricultural information for smallholder farmers, agricultural advisers, and agricultural extension workers. The key values guiding the design are usefulness, relevance, transparency, responsibility, and accountability.

The quantitative evaluation helps connect these values to the technical objective. The initial TF-IDF retrieval baseline achieved an nDCG@5 score of 0.5131 on the training queries, while the

dense retrieval experiment achieved 0.7856. These results show why improving retrieval quality is important: the system needs to place more relevant agricultural documents within the top five positions rather than simply returning documents containing similar words.

The dataset therefore supports experimentation with lexical and semantic retrieval methods while making the limitations of the evaluation visible. The current dataset cannot establish that the system will work equally well for every agricultural community or situation. Before real-world use, further evaluation with appropriate agricultural experts and intended users would be necessary. This provides a basis for remaining accountable to the people the project is intended to support.